Emotion Detection From Tweets

Computational Intelligence - Project

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1. Introduction

Emotion Detection (ED) is a process of determining the human emotion that is being expressed. In this project, the detection was based on text, more particularly tweets. Emotion detection uses Natural Language Processing (NLP), a process that allows machines to understand a text the same way an individual would. ED could also be considered an extension to Sentiment Analysis, which is used to describe the categorization of text as expressing a positive or negative opinion, with ED the goal is not only to identify whether the sentiment is positive or negative, but also to detect what exact emotion it is portraying.

2. Importance of Emotion Detection regarding other fields.

ED has only recently started to be discussed more widely and gained traction within the field of computational intelligence. One of the reasons why the topic has been gaining popularity is ED's wide use in many other fields, one of which being the Economy, where ED found its use in Market Research and Customer Experience. The goal is to use ED to provide optimal service and recommendations to customers, as well as measure their satisfaction based on collected feedback. Customer satisfaction is an important factor for all businesses that rely on customers. Another field that could make good use of ED is media, where ED could help gather audiences' reactions and their reviews, and also help create better content recommendation systems based on understanding viewers' preferences. Analyzing sentiment could also help rid social media spaces of insulting and dangerous content.

One of the most important uses of ED could be in healthcare, where meticulously crafted systems could help monitor the mental health conditions of patients and provide psychological counseling through automated chatbots, helping those needing help and emotional support.

Forensics is another field where ED could prove to be extremely useful. Currently, there is no known method of accurately detecting deception, in the future, machine ED is believed to be able to play that role. Witness testimony analysis as well as assessment of victims' well-being, could also be supported by ED systems.

3. Emotion models

While discussing emotions, it is important to stay true to related discoveries in psychology regarding emotions and their basic categorizations. Regarding that, we can differentiate two different types of emotion models: discrete emotion models and dimensional emotion models.

The discrete model of emotions involves placing emotions into distinct classes or categories:
–The Paul Ekman model distinguishes emotions based on six basic categories. The theory asserts that there exist six fundamental emotions that originate from separate neural systems as a result of how an experiencer perceives a situation, thus emotions are independent. According to this model, the fundamental emotions are happiness, sadness, anger, disgust, surprise, and fear. However, the synergy of these emotions could produce other complex emotions such as guilt, shame, pride, lust, greed, and so on.
–The Robert Plutchik model is a sort of extension of the Ekman model, which adds trust and anticipation to the fundamental emotions, as well as specifies the relation between them, putting them opposite each other as such: joy vs sadness, trust vs disgust, anger vs fear, and surprise vs anticipation.
–Orthony, Clore, and Collins (OCC) model - This model agreed with the Ekman model on emotions arising as a result of how individuals perceived events, but added that they also vary according to their degree of intensity.

The dimensional model presupposes that emotions are not independent and that there exists a relation between them, hence the need to place them in a spatial space:
–Russell presents a circular two-dimensional model called the circumplex of effect. The model proposes two scales: Arousal and Valence domains, with Arousal differentiating emotions by Activations and Deactivations, whereas Valence differentiates emotions by Pleasantness and Unpleasantness. The Circumplex Model of Affect establishes that emotions are not independent but related.
-Plutchik presents a 2-dimensional wheel of emotions that shows Valence on the vertical axis and Arousal on the horizontal axis. The wheel shows emotions in concentric circles with the innermost emotions being derivatives of the eight fundamental emotions, then the eight fundamental emotions, and finally combinations of the primary emotions on the outermost parts of the wheel. The wheel shows how related emotions are according to their positions on the wheel.
–Russell and Mehrabian also present a 3-dimensional emotion model made up of Valence, Arousal, and add Dominance as the third dimension.

Discrete emotion models are often used to represent emotions when designing ED systems, due to their simplicity, although they do not fully exhaust a wider range of emotion classes, their intensities/degrees as compared to the dimensional models.
However, what is important to note is that in existing datasets and works in the field, the emotional categories selected are not always based exclusively on the research done by psychologists regarding emotion theory.

In this project, the model used was a derivative of the Paul Ekman model with 5 fundamental emotions being: sadness (0), joy (1), love (2), anger (3), fear (4), surprise (5). The reason for selecting this model is the lack of accessibility to other datasets where the model used is one of the mentioned above.

4. Basic Exploratory Data Analysis on the chosen dataset

Exploratory Data Analysis (EDA) refers to the process of analyzing and summarizing datasets to gain insights and identify patterns or relationships within the data.

The dataset chosen for this project comes from the Kaggle website which provides a community and a wide range of resources for data science and machine learning enthusiasts.
The name of the dataset: Emotion Dataset for Emotion Recognition Tasks
Link to the dataset:
https://www.kaggle.com/datasets/parulpandey/emotion-dataset?resource=download
The dataset contains three files: training.csv, test.csv, and validation.csv, all of which contain data frames with two columns; one contains the text of a tweet, and the second one a number 0-5, which represents emotion (sadness (0), joy (1), love (2), anger (3), fear (4), surprise (5)).
The picture below shows the first few rows of the training data frame:

	Text	Emotion
1	i didnt feel humiliated	0
2	i can go from feeling so hopeless to so damned...	0
3	im grabbing a minute to post i feel greedy wrong	3
4	i am ever feeling nostalgic about the fireplac...	2
5	i am feeling grouchy	3

Fig 1. First few rows of the data frame used for training the model.

With a few lines of code the numbers in the "Emotion" column were replaced with the emotion behind them, results are shown in Fig 2:

	Text	Emotion
1	i didnt feel humiliated	sadness
2	i can go from feeling so hopeless to so damned...	sadness
3	im grabbing a minute to post i feel greedy wrong	anger
4	i am ever feeling nostalgic about the fireplac...	love
5	i am feeling grouchy	anger

Fig 2. First rows of the data frame with numbers replaced with the emotion they symbolize.

What should also be checked before starting further work is the types of data in the data frame. Fig 3 shows what type of data is contained in both columns:

	column_name	dtype
0	Text	object
1	Emotion	object

Fig 3. Data types in the data frame.

The next figure (Fig 4) shows the number of rows with each emotion:

joy	5362
sadness	4666
anger	2159
fear	1937
love	1304
surprise	572

Fig 4. Number of browns with each emotion.

To better visualize the dataset we can plot a histogram of the emotions in the dataset used for training (Fig 5):

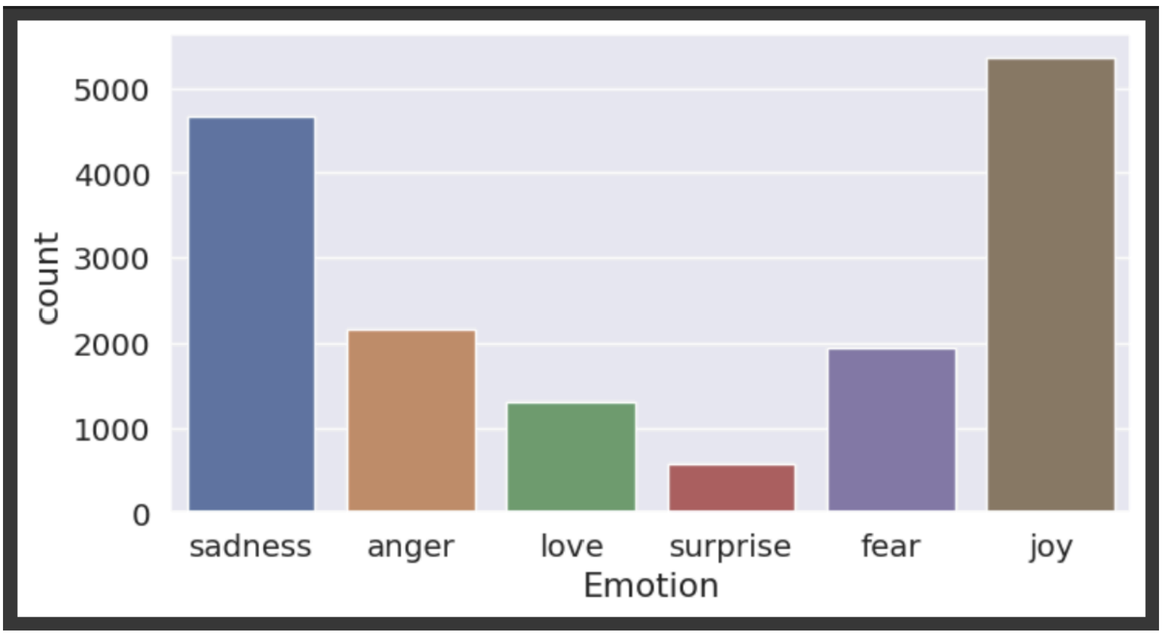

Fig 5. Histogram showing the number of tweets for each emotion

We can see that unfortunately there isn't an equal amount of tweets expressing each emotion, which might cause worse accuracy regarding recognition of those emotions in particular. Later results will show that 'surprise' and 'love' indeed caused the most mistakes in classification. This only highlights the importance of having an equally distributed dataset, with an equal number of representatives for each emotion. That way we can be sure that the dataset itself is not likely to be the reason for lower accuracy scores.

5. Pre-processing

Preprocessing refers to the steps and techniques applied to raw data before it can be used for analysis or modeling. It involves cleaning, transforming, and organizing the data to improve its quality, usability, and compatibility. Often used techniques for pre-processing text data are:
–Tokenization is the process of breaking down either the whole document or paragraph or just one sentence into chunks of words called tokens.
–Normalization involves applying various techniques to remove or reduce variations such as punctuation, capitalization, numerical digits, and special characters. Normalization helps in creating a more uniform representation of text. According to one paper removing punctuation did not affect the accuracy of the tested ED model. Still, according to some other ones punctuation might be an important clue to guessing what emotion is being expressed. For example, users often use elements of punctuation such as exclamation marks when they want to express their strong feelings or an ellipsis to express melancholy, sadness, etc.
–Removing stopwords involves eliminating common words that carry little to no significant meaning in the context of the emotion analysis. Unnecessary words like articles and some prepositions that do not contribute toward emotion recognition and sentiment analysis must be removed. With tweets, there is also a need to consider things like hashtags, URLs, mentions, and emojis. The URLs and mentions are usually removed since they don't bring much intel when it comes to emotions. With hashtags and emojis, it's a bit more complicated, since both can hide useful information regarding the ED. With hashtags, the best solution would be to try to find valid words and separate them (for example "#sohappytobehere" would be replaced with "so happy to be here", which would provide the additional word "happy" which could suggest the emotion of the author). With emojis, the best course of action would be to use a dictionary that replaces the emoji with a word that expresses similar emotions. Of course not all emojis express emotion, in that case, the emoji can be removed. Examples of emojis mapped to emotions are shown in the figure below.

INDICATIVE EXAMPLES OF EMOJI MAPPED TO EMOTIONS

Anger	💢, 🤯, 😤, ...
Anticipation	🔜, ✈️, ...
Disgust	😞, 🤮, ...
Fear	🆘, 😟, ...
Joy	🥇, 😁, 🥰, ...
Sadness	👎, 🙍, 😒, ...
Surprise	😲, 🤨, ...
Trust	⁉️, 👍, ...
No emotion	🐕, 🐈, ☔, ...

Fig 6. Examples of emoji mapped to emotions

–Part-of-Speech (POS) tagging is a way to label different parts of speech in a sentence. These labels can help identify emotional content, as certain parts of speech may be more closely associated with specific emotions. Sentiments and emotions are mostly conveyed by adjectives, For instance, adjectives like "beautiful" or "furious" can provide cues about emotions.
–Stemming and lemmatization. In stemming, words are converted to their root form by truncating suffixes. For example, the terms "argued" become "argue." Lemmatization involves morphological analysis to remove inflectional endings from a token to turn it into the base word lemma. For instance, the term "caught" is converted into "catch".

Another aspect of pre-processing is feature extraction, which is the process of reducing the dimensionality of data by selecting or transforming the original set of features into a new set of features that captures the relevant information. The methods of feature extractions mentioned most in papers regarding text are:
–Bag of Words (BOW) represents text documents as a collection of unique words, disregarding grammar and word order. It creates a "bag" or unordered set of words, along with their frequencies, to capture the presence or absence of words in a document.
–The N-gram method captures the co-occurrence of adjacent words or phrases in the text. N-gram features perform better than the BOW approach as they cover syntactic patterns, including critical information.
–Term frequency-inverse document frequency (TFIDF) represents text in matrix form, where each number quantifies how much information these terms carry in a given text. It is built on the premise that rare terms have much information in the text.
–Word Embeddings represent words in a continuous vector space, capturing semantic relationships between words. In neural network-based word embedding, the words with the same semantics or those related to each other are represented by similar vectors.

Padding is another important factor in pre-processing text, especially of varying lengths. Padding is the process of adding elements to a sequence to make it conform to a certain size or shape. It is commonly used in natural language processing (NLP) tasks to ensure that all sequences have the same length.

In this project, the following pre-processing methods were used: lemmatization, removing stop words, removing numbers, lowercasing the text, removing punctuation, removing extra white space, and removing URLs. The data was also tokenized as well as tfidf was used for ML solutions, for RNN the data was tokenized and padded.

6. Possible Approaches to Emotion Detection

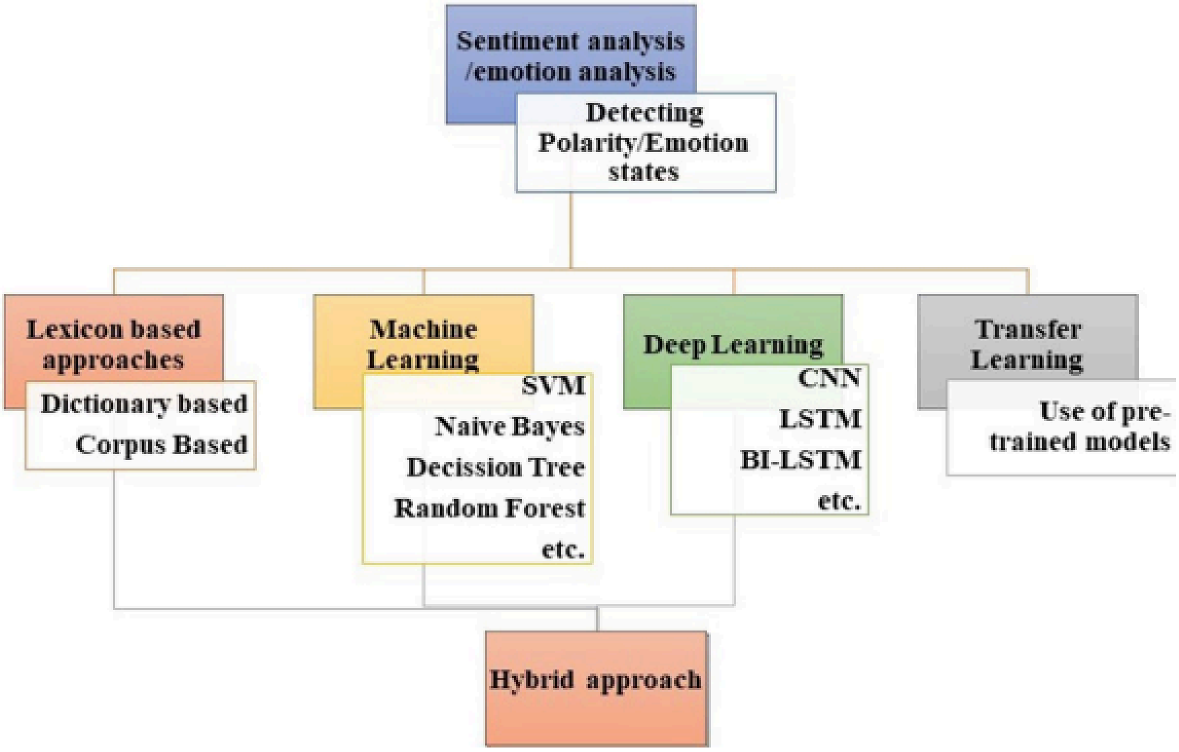

Fig 7. A graph showcasing possible approaches to Sentiment/Emotion Analysis.

Lexicon-based approach (Rule based approach) is a keyword-based search approach that searches for emotion keywords and uses emotion lexicons or dictionaries that associate those words with specific emotions. This approach proposes two possible approaches:
-Dictionary-based approach (Keyword-recognition): In this approach, a pre-defined emotion lexicon or dictionary is used, where words are manually annotated with their corresponding emotions. Dictionary-based approaches provide more control over the emotion labels and their associated words but may be limited by the coverage and specificity of the chosen lexicon.
-Corpus-based approach (lexical affinity (LA)): constructs a lexicon by analyzing a large corpus of text data. The lexicon is built based on statistical analysis and word frequencies within the data. Words that frequently co-occur with certain emotions are identified and

associated with those emotions. The corpus-based approach provides more flexibility but potentially lacks in capturing specific or domain-specific emotions. Popular lexicons are Word-Net-Afect and NRC word–emotion lexicon. Researchers often employ a combination of these approaches to enhance accuracy.

The Machine Learning (ML) approach is based on training models on labeled datasets where emotions are manually annotated. These models learn to recognize patterns and features that indicate different emotions in text. There are a lot of models used in ED, but the ones mentioned most often are Support Vector Machines (SVM), Naive Bayes, Decision Trees, Random forests, and Logistic Regression.

The Deep Learning approach utilizes deep neural networks to automatically extract and learn complex features from the text.

Hybrid approaches combine multiple techniques, such as using lexicon-based methods as features in machine learning models or incorporating deep learning architectures into sentiment analysis pipelines.

Transfer learning is a machine learning technique where knowledge gained from training a model on one task is applied to a different but related task. It involves leveraging pre-trained models that have been trained on large datasets to extract useful features or as a starting point for training new models on a target task, enabling faster and more accurate learning with limited data.

7. Used ML models

In order to have a scale to compare the results of the final model, ML models were used to access what accuracy can be achieved by simpler models.

The table below shows a comparison of the accuracies achieved with a few different ML models:

	Model	Accuracy
0	Random Forest	0.876
1	Logistic Regression	0.870
2	Support Vector Machine	0.870
3	MLP Classifier	0.870
4	Decision Tree	0.810

Fig 8. Accuracy of chosen ML models.

To better analyze the results we can also present them in the form of a confusion matrix.

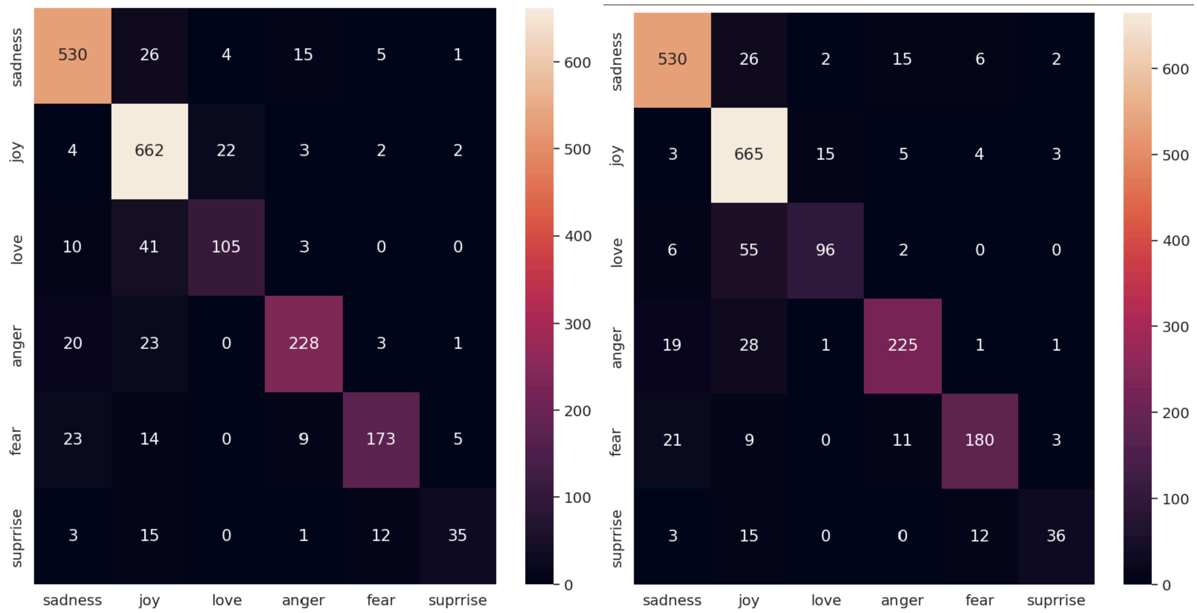

Fig 9. Confusion matrix for the Logistic Regression model (left) and SVC (right).

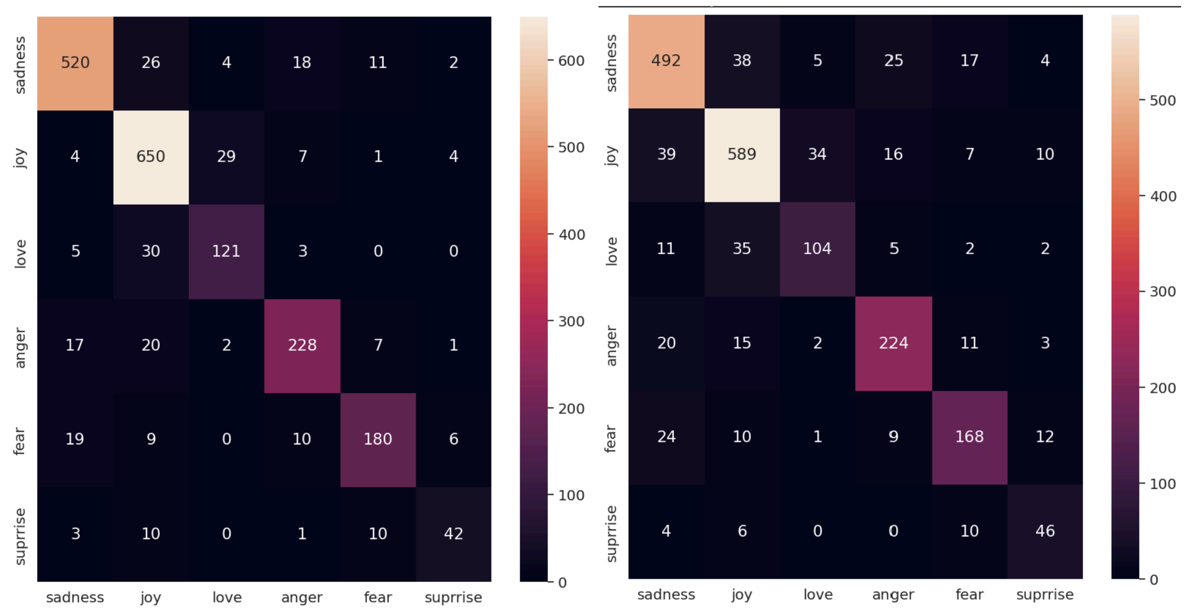

Fig 10. Confusion matrix for MLP Classifier (left) and Decision Tree (right).

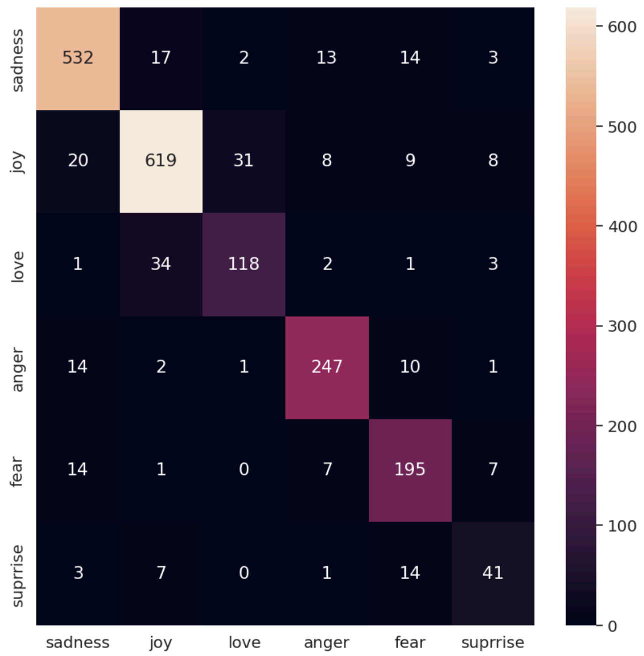

Fig 11. Confusion matrix for Random Forest.

From the results we can see that the Random Forest algorithm did best, achieving an accuracy of 87.6%. The confusion matrixes give us more information on the categories where the model struggled the most. Surprise and love were the emotions that had the least accuracy, especially surprise. After looking at the distribution of the training data a conclusion can be made that a lower quantity of data with a certain tag can cause that tag to be least recognized and often misclassified. This showcases how important it is to make the training data uniform.

8. RNN

In this project, RNN (Recurrent Neural Network) with a combination of bidirectional LSTM (Long Short-Term Memory) layers is used. Bi-LSTM processes input sequences in both forward and backward directions simultaneously. It leverages information from both past and future contexts, making it effective for tasks that require understanding context in both directions, such as sentiment/emotion analysis or named entity recognition. The dropout layers are used as well, for regularization to prevent overfitting. In the figure below the structure of the model is shown.

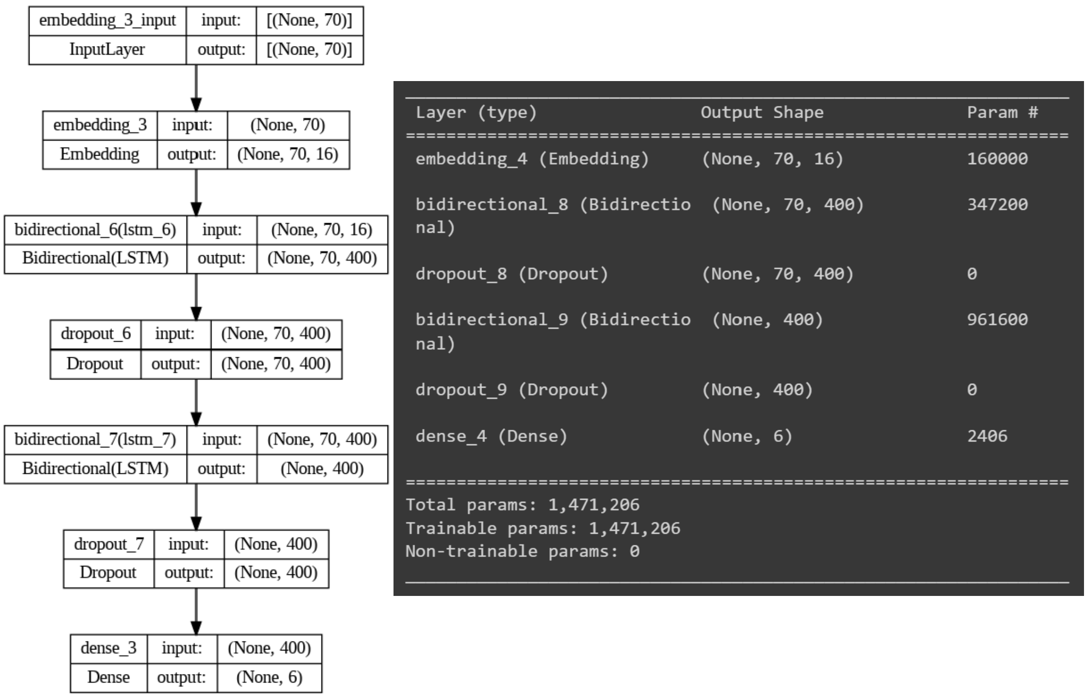

Fig 12. Structure of the model.

With this model, the best-achieved accuracy was 0.92, which is a better score than any of the ML approaches was able to achieve.

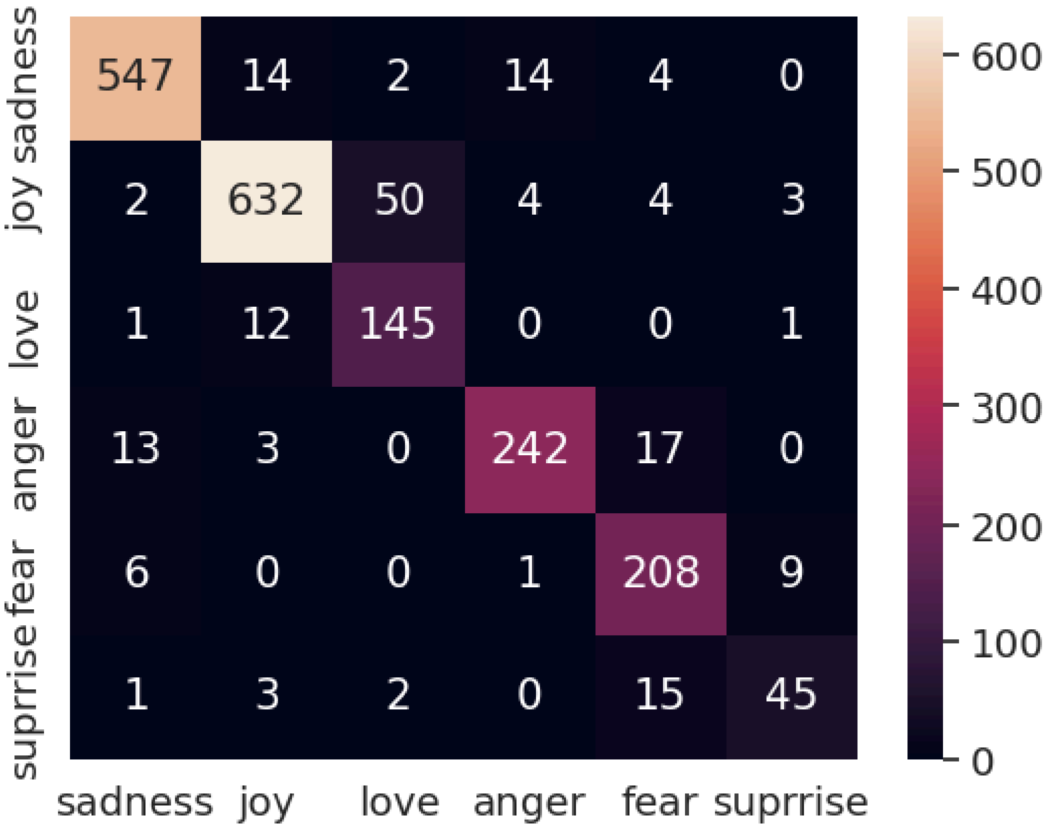

Fig 13. Confusion matrix for the RNN.

To better observe the results a classification report can also be created, that will present values for parameters like precision, recall, f1-score, and support.

Classification Report:

	precision	recall	f1-score	support
sadness	0.97	0.95	0.96	581
joy	0.95	0.91	0.93	695
love	0.69	0.89	0.78	159
anger	0.94	0.87	0.90	275
fear	0.85	0.95	0.90	224
suprrise	0.80	0.73	0.76	66
accuracy			0.91	2000
macro avg	0.87	0.88	0.87	2000
weighted avg	0.92	0.91	0.91	2000

Fig 13. Classification Report

In this model, the ‘surprise’ and ‘love’ tags were also the ones that caused the most problems in the classification, but the RNN has more good classifications for ‘love’ than any ML approach and was mostly confused with ‘joy’, which at least is in a way a similar emotion. Further analysis of the model shows that most of the misclassifications are at least in the same ‘group’ of emotions, for example, fear was confused with sadness and surprise and anger with sadness and fear, all of which would be considered negative emotions. Mistakes where the system labels a negative emotion as a positive one are rare, which is a good indication that the system is at least able to properly recognize the general sentiment of the tweet.

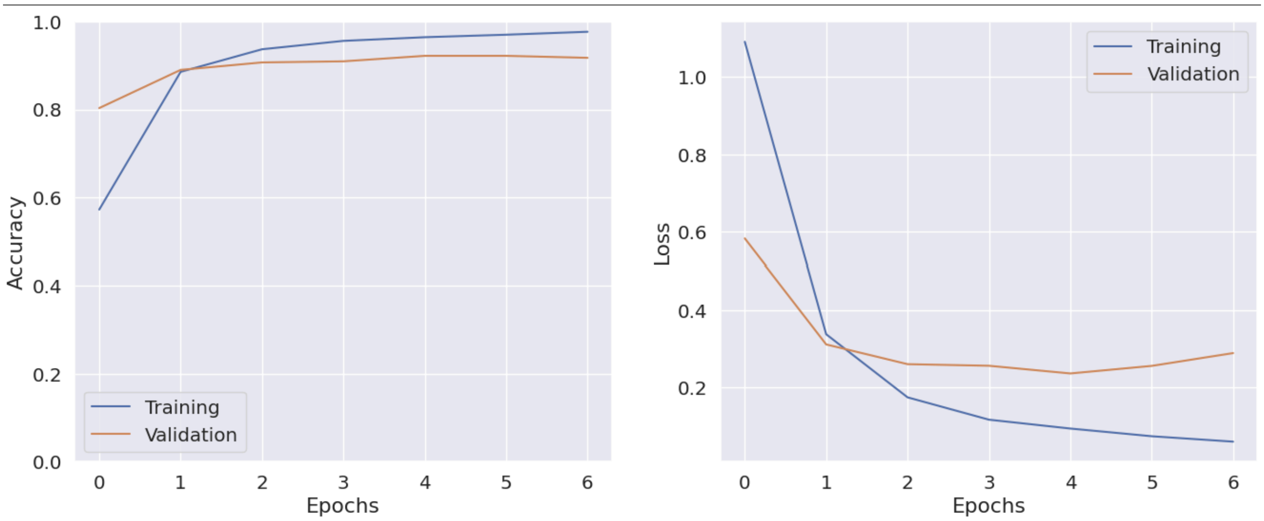

Fig 14. Plots of accuracy in Epochs, as well as Loss over epochs.

If we look at the figure on the right (Fig 14) we can see that the model has a slight problem with overfitting. It is difficult to say if the model has reached a stable state from this figure, but while performing the task on a bigger number of epochs it proved that the model shows no improvement in the further epochs.

Looking at other works regarding the dataset used in the project, similar results could be found, some lower, some higher (0.932 accuracy). The model that achieved higher accuracy had much more layers and was much more complicated, but the accuracy was not improved by a large margin. In one of the papers LSTM model had an accuracy of 91.90% and it was the highest among other tried in the paper models. The model created in the project shows similar results to the existing ones.

9. Ways to possibly improve

To improve the accuracy more work could be done regarding the pre-processing, for example, instead of removing emojis, the dictionary could be used to replace them with the emotion they represent. Making changes to the model itself could also help improve the accuracy of the RNN model.
Another thing worth working on is uniforming the dataset, making sure the system has enough information on each tag to properly recognize each emotion.
Different optimizers could also be used to try and improve the performance of the model.

10. Difficulties in Emotion Detection

Text data is not the easiest to work with regarding ED. Below are mentioned the most prominent difficulties that are often met during working on an ED system, most of which caused an issue in this project as well.
-Limited availability of training datasets. The issue here is not only the limited availability but also a high number of unreliable datasets where either the emotion categories are not compliant with research in psychology on emotions or the classification done in the dataset is of questionable quality.
-Most of the resources are available in the English language. In this project that wasn't an issue but if there was an attempt of doing the same work with Tweets in Polish for example, it might have been difficult to find a dataset fitting enough, if any.
-Emotion interpretation by humans is biased. The understanding and interpretation of emotions is a skill that people develop through their environment, the other people they communicate with, and their social and cultural interactions. Thus the categorizing done by humans might not always be perfect, and up to debate.
-Continuous evolution of new words due to language dynamics, spelling mistakes, new slang, WEB slang, and incorrect use of grammar. Each of the mentioned issues makes it difficult to train a universal system that would stay relevant. Mistakes in spelling and grammar can also cause difficulty for the system to accurately recognize words that would allow it to detect emotion.

-Sarcasm and irony. This point seems pretty self-explanatory. Especially in written form sarcasm and irony are often close to impossible to recognize.
-Comparative sentences. For example, consider two sentences 'Phone A is worse than Phone B' and 'Phone B is worse than Phone A.' The word ’worse’ in both sentences will signify negative polarity, but these two sentences oppose each other.
-Multiple meanings of words based on the overall context
-Text objectivity and originality, topic diversity (a single social media post can include multiple topics, and express opinions of different intensity, at the same time), restricted information contained in a short text, texts may not portray peculiar cues to emotions (text has no info from body language, facial expressions, tone of voice, etc.), lack of contextual information. All are issues difficult to get over for a machine.
-Statements conveying several emotions
-Traditional NLP systems have been trained using large passages of text that have only one author. Social media posts have limited characters and are produced by multiple individuals.
All the difficulties mentioned above prove ED to be a challenging task, that in an uncontrolled environment could prove to be unsuccessful.

11. Lessons learned

This project allowed for the systematization of the knowledge gained during the lectures, as well as gaining insight into Emotion Detection, emotion theory as well as handling text data, especially tweets, which differ from the usual long-form text in many aspects. Preparation for the project helped discover different approaches to Emotion Detection and its importance in different fields and directions in which ED could extend and improve in the future.

12. Possible issues to focus on in the future

What’s worth mentioning is the direction of further work that can be done regarding improving the ED systems. Below are a few interesting directions to take ED that was mentioned in the literature:
-Detecting multiple emotions and sentiments from the same text;
-Realizing the cause behind an emotion or sentiment;
-Detecting the personality of a person;
-Relating emotion and sentiment with different social and individual parameters;
-Detecting the mood of an individual or public in general;
-Predicting the action of a person based on their emotion or sentiment.

13. Conclusions

In conclusion, emotion analysis from text offers powerful insights into understanding and interpreting human emotions at scale. As technology continues to advance, emotion analysis holds immense potential for unlocking new opportunities and enhancing our understanding of human emotions in the digital age.

Despite so many existing works described in the literature, the accuracy of emotion detection from a text when the emotion is implied, not plainly expressed is not good. Detecting sentiments from texts have better accuracy than detecting emotions, detecting sarcasm correctly still needs huge contributions.

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